



Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT

JEE-MAIN

Date: 04-01-2026

Time: 09.00Am to 12.00Pm

GTM-09

Max. Marks: 300

KEY SHEET

MATHEMATICS

1	3	2	2	3	3	4	1	5	2
6	1	7	1	8	4	9	1	10	3
11	3	12	1	13	4	14	2	15	3
16	1	17	2	18	1	19	3	20	1
21	13	22	3	23	3	24	2	25	3

PHYSICS

26	1	27	3	28	1	29	4	30	4
31	2	32	1	33	2	34	3	35	1
36	3	37	3	38	2	39	1	40	4
41	4	42	2	43	3	44	3	45	2
46	21	47	2	48	18	49	6	50	5

CHEMISTRY

51	3	52	4	53	2	54	1	55	2
56	2	57	4	58	3	59	2	60	4
61	1	62	1	63	4	64			



SOLUTION

MATHEMATICS

1. $\vec{AC} = \vec{AB} + \vec{BC}$

$$\vec{BC} = \frac{\vec{u}}{|\vec{u}|} + \frac{\vec{v}}{|\vec{v}|} \Rightarrow \vec{AB} \cdot \vec{BC} = \left(\frac{\vec{u}}{|\vec{u}|} - \frac{\vec{v}}{|\vec{v}|} \right) \cdot \left(\frac{\vec{u}}{|\vec{u}|} + \frac{\vec{v}}{|\vec{v}|} \right)$$

$$= \frac{|\vec{u}|^2}{|\vec{u}|^2} - \frac{|\vec{v}|^2}{|\vec{v}|^2} = 0$$

$$\angle B = \frac{\pi^c}{2} \text{ Hence } \sum \cos 2A + 1 = 0$$

2.

$$\frac{x}{1} = \frac{y-2}{1} = \frac{z+4}{-1}$$

3. Let $A_1, A_2, \dots, A_{2n}$ be the vertices of regular polygon of n sides

Now select A_1 and the remaining vertices from the right hand side i.e., $n-1_{C_2}$.

Similarly on the left hand side also.

($\because$ circumcentre always lies outside the obtuse triangle)

$$\text{So required probability} = \frac{2 \times 2n \times {}^{n-1}C_2}{2n_{C_3}}$$

$$\text{(Here } 2n \text{ is even) or when } n \text{ is odd } \frac{2 \left(\frac{n-1}{2} \right) C_2 \cdot n}{2 \left(n_{C_3} \right)}$$

4. R be $(2 \cos \theta, 3 \sin \theta)$, OP perpendicular to OR

$$\therefore \frac{3}{2} \tan \theta \times \frac{3}{\sqrt{3}} = -1 \Rightarrow \tan \theta = \frac{-2\sqrt{3}}{9}, \text{ required } \frac{1}{4} \left(\frac{1}{OP^2} + \frac{1}{OR^2} \right) = \frac{13}{144}$$

5. Given $\int \tan^{-1} x \cdot \frac{1}{x(\ln x)} + \frac{1}{1+x^2} \cdot \ln(\ln x) \therefore f(x) = (\tan^{-1} x) \ln(\ln x)$

6. By differentiating wrt x , we get $(1+x^2)f(x) = x \rightarrow f(x)$ is odd

$$\text{Required integral} = \int_{-\pi/4}^{\pi/4} \sec^2 x dx = 2$$

7. $D = p^2 - 5p$. If $p = 9$, then D is perfect square



9. $S = 0$ as, $\sum_{r=0}^n n_{c_r} (-1)^r$ (2026) = 0

10. The given equation is $(x-1)(x-2)(x-3) = 0$

11. $\cos\left(2\sin^{-1}\frac{1}{\sqrt{5}}\right) = \frac{3}{5}, \sin^{-1}\left(\sin\frac{2\pi}{3}\right) = \frac{\pi}{3}$

$$\cos^{-1}\left(\cos\frac{7\pi}{6}\right) = 2\pi - \frac{7\pi}{6}$$

12. use L'Hospitals rule

13. $|\text{adj}(\text{adj } A)| = |A|^{n-1}$

14. $f''(1) = 3 - 2a + b = af''(2) = 12 - 2a = b, f'''(3) = 6 \therefore a = 3, b = 6$

15. Conceptual

16. $\int e^y dy = \int 2x dx = f(x) = \ln(x^2 + c)$
 $f'(1) = 1 \Rightarrow C = 1$

17. Conceptual

18. Conceptual

19. Conceptual

20. The number of distinct reflexive relations defined on set $A = 2^{n(n-1)}$ where $|A| = n$

21. Conceptual

22. $(\sin \alpha \sin \beta \sin \gamma) = \frac{4^3 (\sin \alpha \sin \beta \sin \gamma)^2}{(1 + 4 \sin^2 \alpha)(1 + 4 \sin^2 \beta)(1 + 4 \sin^2 \gamma)}$

$$\Rightarrow \text{either } \sin \alpha \sin \beta \sin \gamma = 0 \Rightarrow \sin \alpha = \sin \beta = \sin \gamma = 0$$

OR

$$(1 + 4 \sin^2 \alpha)(1 + 4 \sin^2 \beta)(1 + 4 \sin^2 \gamma) = (4 \sin \alpha)(4 \sin \beta)(4 \sin \gamma)$$

$$\Rightarrow \sin \alpha = 1/2 = \sin \beta = \sin \gamma$$

$$\cos \alpha = \pm \frac{\sqrt{3}}{2} = \cos \beta = \cos \gamma$$

$$\text{max value} = \frac{\left(\left(\frac{1}{2}\right) + \left(\frac{1}{2}\right) + \left(\frac{1}{2}\right)\right)^2}{\left(\frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2}\right)^2} = 3.$$

23. $(\sin x + 3)(\sin x - (x-1)^2) = 0$



$$25. \quad \mu = \frac{(40 \times 12.5 - 10 \times 15)}{25} = 14 \sigma^2 = 2 + \frac{15 \times 25}{40 \times 40} (4)^2 = 5.75$$

PHYSICS

$$26. \quad K_1 = \frac{hc}{\lambda_1} - \phi_0 \quad \dots(1)$$

$$K_2 = \frac{hc}{\lambda_2} - \phi_0 \quad \dots(2)$$

$$\text{From eqn. (1)} \quad \frac{hc}{\lambda_1} = K_1 + \phi_0 \quad \text{or } hc = \lambda_1 (K_1 + \phi_0) \quad \dots(3)$$

Put eqn. (3) in eqn. (2)

$$K_2 = \frac{\lambda_1}{\lambda_2} (K_1 + \phi_0) - \phi_0 = \frac{\lambda_1}{\lambda_2} K_1 + \phi_0 \left(\frac{\lambda_1}{\lambda_2} - 1 \right)$$

$$\text{Or } \phi_0 \left(\frac{\lambda_1 - \lambda_2}{\lambda_2} \right) = K_2 - \frac{\lambda_1}{\lambda_2} K_1 = \frac{\lambda_2 K_2 - \lambda_1 K_1}{\lambda_2} \quad \text{Or } \phi_0 = \frac{\lambda_2 K_2 - \lambda_1 K_1}{\lambda_2}$$

$$\text{Or } \phi_0 = \frac{\lambda_2 K_2 - \lambda_1 K_1}{(\lambda_1 - \lambda_2)}$$

$$27. \quad \text{Here } C = A, B \text{ and } D = \bar{A}B$$

$$E = C + D = (A.B) + (\bar{A}.B)$$

Explanation. The truth table of this arrangement of gates can be given below:

A	B	$\bar{A}$	$C = A.B$	$D = \bar{A}.B$	$E = (C + D)$
0	0	1	0	0	0
0	1	1	0	1	1
1	0	0	0	0	0
1	1	0	1	0	1

$$28. \quad \text{As, force, } F = \frac{e^2}{4\pi\epsilon_0 r^2} \Rightarrow \frac{e^2}{4\pi\epsilon_0} = r^2.F$$

Putting dimensions of r and F, we get.



$$\Rightarrow \left[\frac{e^2}{4\pi\epsilon_0} \right] = [ML^3T^{-2}] \dots\dots\dots (I)$$

Also, force, $F = \frac{Gm^2}{r^2} \Rightarrow [G] = [M^{-1}L^3T^{-2}] \dots\dots (ii)$

and $\left[\frac{1}{c^2} \right] = \left[\frac{1}{L^2T^{-2}} \right] = [L^{-2}T^2] \dots\dots (iii)$

Now, checking options wise
$$= \frac{1}{c^2} \left(\frac{Ge^2}{4\pi\epsilon_0} \right)^{1/2} = [L^{02}T^2] [L^6T^{-4}I^{1/2}] = [L]$$

29. Conceptual

30. $E = \frac{1}{2} mA^2 \omega^2$ i.e. $E \propto (A\omega)^2$ or $(A_1\omega_1)^2 = (A_2\omega_2)^2$

31. Conceptual

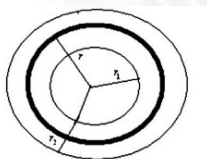
32. Conceptual

33. Conceptual

34. 1) $\chi \propto \frac{1}{T}$ 2) $B = 0$ $r < R$, $B \propto \frac{1}{r}$ $r > R$

3) $B \propto r$, $r < R$, $B \propto \frac{1}{r}$ $r > R$ 4) $B = \frac{\mu_0}{4\pi} \frac{I\pi R^2}{(R^2 + r^2)^{3/2}}$

35.



$$H = \frac{-KAdT}{dr} \quad H = \frac{-K4\pi r^2}{dr} dT$$

$$\frac{dr}{r^2} = \frac{-4\pi K}{H} dT \quad \int_{r_1}^{r_2} \frac{dr}{r^2} = \frac{-4\pi K}{H} \int_{T_1}^{T_2} dT$$

$$\Rightarrow \left[\frac{1}{r_1} - \frac{1}{r_2} \right] = \frac{-4\pi K}{H} [T_2 - T_1] = H \propto \frac{r_1 r_2}{r_2 - r_1}$$

36. $F_2 \Rightarrow (\rho g R)(\pi R^2)$



$$F_2 \Rightarrow \rho \left(\pi R^3 - \frac{2}{3} \pi R^3 \right) g$$

$$F_1 = \rho \left(\frac{\pi R^3}{3} \right) g$$

$$\left[\frac{F_1}{F_2} = \frac{1}{3} \right]$$

37. If A moves down by a_1 w.r.t B and B moves to the right by a_2 then the constraint relation gives $a_1 = a_2 = a$.

For A : $N = ma; mg - T = ma$

For B : $T - N = ma \Rightarrow T = 2ma$

$$\Rightarrow a = \frac{g}{3} \quad \Rightarrow \vec{a}_{AO} = \vec{a}_{BO} + \vec{a}_{AB} = \frac{g}{3} \hat{i} - \frac{g}{3} \hat{j}$$

38. Since $ML = Pt \Rightarrow L = \frac{Pt}{M}$

39. $(2v_0) \times e \geq v_0 \times \sqrt{3} \Rightarrow e \geq \frac{\sqrt{3}}{2}$

40. As specific resistance does not depend on dimension of wire so, it will not change.

41. $\vec{S} = \vec{ut} + \frac{1}{2} \vec{at}^2$

42. Detector will receive maximum light when the image of the source gets formed at the detector itself. For any general time,

Position of source from lens = $-[10\text{cm} - (3\text{cm/s} - 1\text{cm/s})t] = -(10\text{cm} - (2\text{cm/s})t)$

Position of detector from lens = $(3\text{cm/s} - 1\text{cm/s})t = (2\text{cm/s})t$

Using the lens formula, $\frac{1}{0.9} = +\frac{1}{(10-2t)} + \frac{1}{2t}$

$(10-2t)2t = 9 \quad t = 0.5 \text{ sec and } t = 4.5 \text{ sec.}$

$4t^2 - 20t + 9 = 0$

43. $Q = \Delta U + W = \Delta U + \eta Q \Rightarrow (1-\eta)Q = \Delta U$

$(1-\eta) \frac{Q}{n\Delta T} = \frac{\Delta U}{n\Delta T} \quad (1-\eta).C = \frac{5}{2}R$

44. Key: C

45. Conceptual



46. n is number of polarides then $I_f = \frac{I_0}{2} (\cos^2 \theta)^{n-1}$

47. $B_y = \frac{\mu_0}{4\pi} \frac{2\pi / (2r^2)}{[(2r)^2 + (2d)^2]^{3/2}}, B_x = \frac{\mu_0}{4\pi} \frac{2\pi / r^2}{[r^2 + d^2]^{3/2}}$

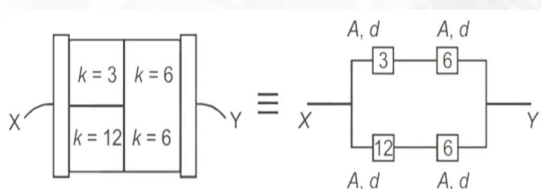
Solve to get $\frac{B_x}{B_y} = 2$

48. $E_1 = \frac{1}{2} Kx^2 = 2 \quad x = \frac{2}{\sqrt{K}}$

$E_2 = \frac{1}{2} Ky^2 = 8 \Rightarrow y = \frac{4}{\sqrt{K}}$

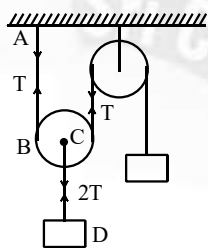
$E = \frac{1}{2} K(x+y)^2 = \frac{1}{2} K \left[\frac{2}{\sqrt{K}} + \frac{4}{\sqrt{K}} \right]^2 \quad E = 18 J$

49.



Equivalent capacitance of the circuit $\frac{k\epsilon_0(2A)}{(2d)} = \frac{3 \times 6}{3+6} \cdot \frac{\epsilon_0 A}{d} + \frac{6 \times 12}{6+12} \cdot \frac{\epsilon_0 A}{d} \Rightarrow k = 2 + 4 = 6$

50. $V_1 = \sqrt{\frac{T}{\mu}}$



$V_2 = \sqrt{\frac{2\%}{\mu}}, \frac{V_1}{V_2} = \frac{1}{\sqrt{2}}$



CHEMISTRY

$$51. \quad \frac{1}{2}mv_{\max}^2 = \frac{hc}{\lambda} - \frac{hc}{\lambda_0} \Rightarrow v_{\max} = \sqrt{\frac{2hc}{m} \left(\frac{\lambda_0 - \lambda}{\lambda_0 \lambda} \right)}$$

52. Non polar compound is eluted early.

53. IP of Nitrogen is more than Oxygen.

54. **HClO is weak acid . So ClO^- is strong base .**

55. Reimer tiemann reaction followed by cyanohydrin formation.

56. Elimination followed by addition reaction

57. $\text{Al}_2 - \text{Cl}_6$ does not possess $\text{Al} - \text{Al}$ bonds.

$$58. \quad (61.9 + 76.3) = \frac{138.2 \times 10^{-4} \times 1000}{S} \quad \therefore S = 10^{-3} M.$$

$$2^x = 8$$

$$59. \quad 2^x = 2^3$$

$$x = 3$$

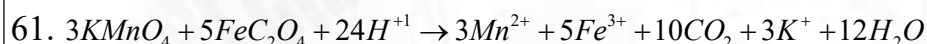
$$t = 4 \text{ min}; a - 3x = x$$

$$\Rightarrow 4x = a \Rightarrow x = \frac{a}{4}$$

$\therefore$ At 4 min 75% of first order is completed.

$$\therefore t_{75\%} = \frac{2t_1}{2} \Rightarrow \frac{t_1}{2} = 2 \text{ min.}$$

60. NH_3^+ -acts as both ortho, para and meta directing group in the presence of acid due to salt formation.



$$\text{No of } e^- \text{ ratio} = \frac{3}{5}$$

62. Lanthanoid contraction.

63. In liquid phase

$$P_T = P_A^0 \chi_A + P_B^0 \chi_B = 120 \times 0.6 + 50 \times 0.4 \\ = 72 + 20 = 92 \text{ mmHg}$$

$$\text{In vapour phase } \chi_1 = \frac{P_{\text{Toluene}}}{P_{\text{Total}}} = \frac{$$



68. $S = \frac{7.8}{78} \times 10^{-4} \text{ mol/l} = 10^{-5} \text{ mol/l}$

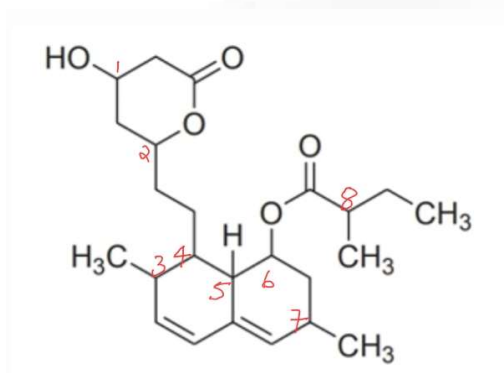
$K_{sp} = 4S^3 = 4 \times 10^{-15} \therefore \log_{10} K_{sp} = \log 4 - 15 = -14.4$

69. $H_{(C)}^+ \rightleftharpoons H_{(A)}^+$

$E_{cell} = -\frac{0.06}{1} \log \frac{0.001}{0.2} = 0.06 \log 2 \times 10^2 = 0.06(0.3 + 2) = 0.06 \times 2.3 = 138V$

70. Benzylamine, the lone pair on nitrogen is free

71.



72. FORMIC ACID, BENZALDEHYDE, ACETALDEHYDE

73. For each 2 CHOH – one mole of HCHO is consumed.

74. 2 cis and 1 trans possible

75. A decapeptide has nine peptide (amide) linkages, to break each peptide linkages one water molecule is used.

Hence, total mass of hydrolysis product = $796 + 18 \times 9 = 958$

$\Rightarrow$ Mass of glycine in hydrolysis product = $\frac{958 \times 47}{100} = 450$.

$\Rightarrow$ Number of glycine molecule in one molecule of decapeptide = $\frac{450}{75} = 6$.